IMAGE PROCESSING TECHNIQUES – EXTENDED CLASS ACTIVITIES WORKSHEET

UG Level | Duration: 10 Minutes per Activity

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Date: 07/09/2026

Activity 2: Image Processing Pipeline

What is the purpose of pre-processing?

Pre-processing is a fundamental step in image processing that prepares an image for further analysis and processing. The main purpose of pre-processing is to improve the overall quality of the image by removing unwanted noise, distortions, and unnecessary information. It may involve techniques such as noise reduction, image smoothing, sharpening, contrast enhancement, brightness adjustment, and resizing. These techniques make important features of the image more clear and distinguishable. Proper pre-processing also helps overcome problems caused by poor lighting, low image quality, or variations during image acquisition. As a result, the processed image becomes easier for a computer to analyze, which improves the accuracy and reliability of subsequent steps such as segmentation, feature extraction, and classification.